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(72) Inventors: **Ramasamy VENKATRAGAVAN**, Thiruvallur (IN); **Erugu BALAJIAH**, Thiruvallur (IN); **Paramanandam SENTHILKUMARAN**, Thiruvallur (IN); **Guttha JAYAPRASAD**, Thiruvallur (IN); **Manniyam Kailasam SELVAM**, Thiruvallur (IN); **Noti KRISHNAREDDY**, Thiruvallur (IN); **Sorakka Pichandi PARTHIPAN**, Thiruvallur (IN); **Murugan ARUNKUMAR**, Thiruvallur (IN); **Palaniswamy PREMKUMAR**, Thiruvallur (IN); **Salamuthu KALIRAJ**, Thiruvallur (IN); **Shanmugam ARUMUGAM**, Thiruvallur (IN); **Singavarapu Ajay MADHUKAR**, Thiruvallur (IN); **Raman SILAMBARASAN**, Thiruvallur (IN); **Saminathan KARTHICK**, Thiruvallur (IN); **Kudumudi Jayaraman PARTHIBAN**, Thiruvallur (IN); **Ravi SILAMBARASAN**, Thiruvallur (IN)

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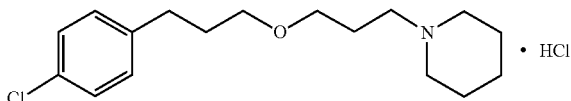
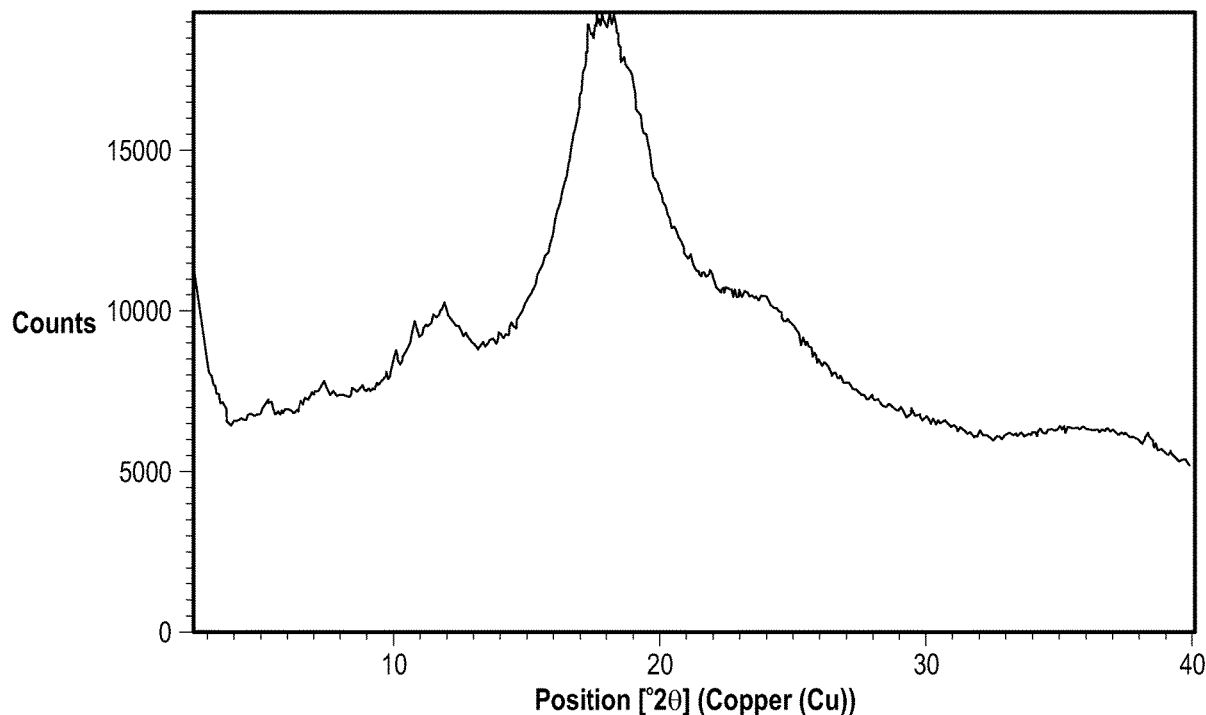
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(57) **ABSTRACT**

The present disclosure relates to a process for preparing pitolisant hydrochloride of Formula-(I) and solid-state forms thereof.

Formula-(I)

(73) Assignee: **Nuray Chemicals Private Limited, Thiruvallur (IN)**

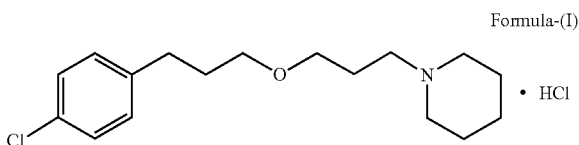
PROCESS FOR PREPARING PITOLISANT HYDROCHLORIDE AND SOLID-STATE FORMS THEREOF

RELATED APPLICATIONS

[0001] Priority is claimed to Indian Patent Application Nos. 202141025328, filed on Jun. 7, 2021; 202141035609 filed on Aug. 6, 2021; 202141046743 filed on Oct. 13, 2021; and 202241009416 filed on Feb. 22, 2022.

FIELD OF THE INVENTION

[0002] The present disclosure relates to an improved process for preparing pitolisant hydrochloride of Formula-(I) and its solid-state forms thereof.

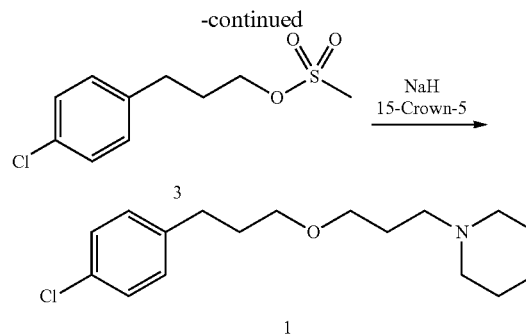
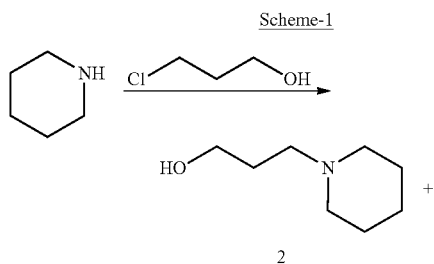


BACKGROUND

[0003] Pitolisant hydrochloride is chemically known as 1-{3-[3-(4-chlorophenyl)propoxy]propyl}piperidine hydrochloride and it is an antagonist/inverse agonist of the histamine-3 (H3) receptor. Pitolisant hydrochloride is approved for the treatment of excessive daytime sleepiness (EDS) in adult patients with narcolepsy. Pitolisant hydrochloride is a white or almost white crystalline powder, and it is soluble in water, ethanol, and methylene chloride and practically insoluble in cyclohexane.

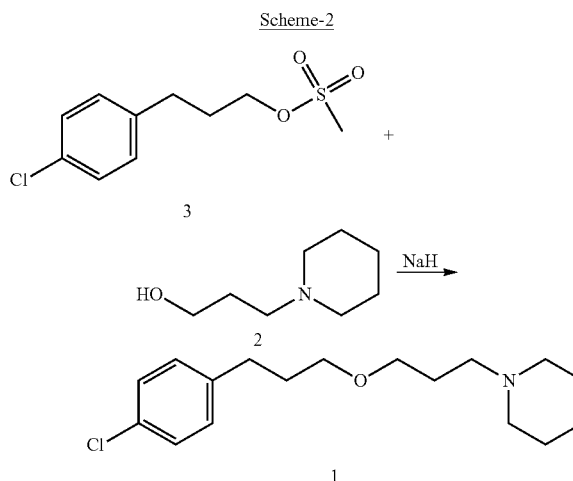
[0004] Schwartz discloses a process for preparing pitolisant and its pharmaceutically acceptable salts such as hydrochloride, hydrobromide, hydrogen maleate, and hydrogen oxalate.

[0005] Meier discloses a three step process for preparing pitolisant hydrochloride starting with piperidine and 3-chloropropanol, to give the intermediate 3-(piperidin-1-yl)propan-1-ol (2), which is reacted with 3-(4-chlorophenyl)propyl methanesulfonate (3) in the presence of sodium hydride and 15-crown-5 to give pitolisant (1) which is finally reacted with gaseous hydrochloric acid to obtain pitolisant hydrochloride, as shown in Scheme-1.

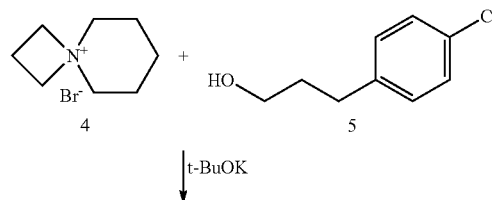


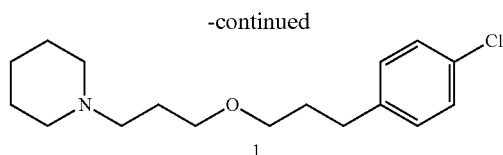
[0006] Raga discloses a crystalline form of pitolisant monohydrochloride having a moisture content of 2.7 to 6.5% and its preparation thereof. Raga discloses pitolisant oxalate and pitolisant maleate and their respective solubilities in water. Raga discloses that pitolisant hydrochloride exhibits different properties (e.g., aqueous solubility and stability) when compared to pitolisant oxalate and pitolisant maleate.

[0007] Sallarés discloses a process for preparing pitolisant (1) by forming the sodium salt of 3-(piperidin-1-yl)propan-1-ol (2) by reaction with sodium hydride (NaH), and then reacting the sodium salt with 3-(4-chlorophenyl)propyl methanesulfonate (3), as shown in Scheme-2.



[0008] Colombano discloses a process for preparing pitolisant (1) by reacting 4-azaspiro[3.5]nonan-4-ium bromide (4) with 3-(4-chlorophenyl)propan-1-ol (5).

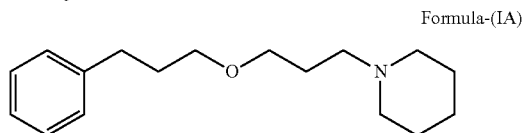
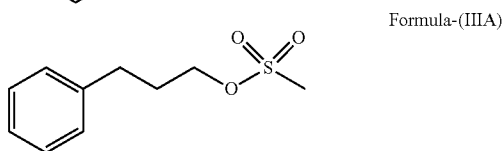
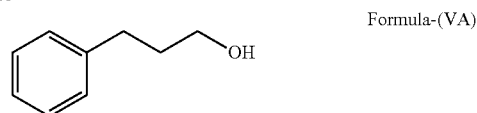
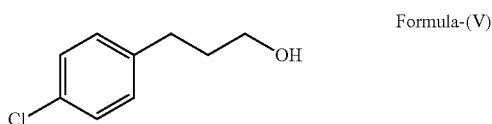




[0009] Sefinovic discloses a hydrogen fumarate salt of pitolisant and a process for its preparation.

[0010] The prior manufacturing processes for pitolisant utilize sodium hydride (NaH), which is known to be a hazardous and pyrophoric reagent. They have their own handling issues because of its pyrophoric nature and hence these processes are not suitable for a commercial scale.

[0011] The reported methods for preparing the compound of Formula-(V) result in the formation of deschloro compound of Formula-(VA) with desired compound of Formula-(V). The deschloro compound having almost same property of compound of Formula-(V), it is cumbersome to remove this impurity from compound of Formula-(V) and its carryover to the next steps results in deschloro impurities selected from deschloro mesyl compound of Formula-(IIIA) and deschloro pitolisant of Formula-(IA). The removal of these impurities is tedious and needs further purification steps.



[0012] Hence, there is a need for a commercially suitable and non-hazardous process for preparing pitolisant hydrochloride which does not contain deschloro impurities. The following disclosure is the result of an extensive investigation for an improved process for preparing pitolisant hydrochloride using non-hazardous bases, as well as the discovery of novel solid-state forms of pitolisant hydrochloride of Formula-(I).

OBJECTIVES OF THE INVENTION

[0013] The main objective of the present disclosure is to provide a non-hazardous process for preparing pitolisant and its pharmaceutically acceptable salts thereof.

[0014] Another objective of the present disclosure is to provide new solid-state forms of pitolisant and its pharmaceutically acceptable salts.

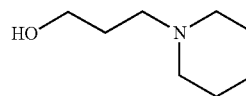
[0015] Still another objective of the present disclosure is to provide an improved process for preparing 3-(4-chlorophenyl) propanol of Formula-(V), a useful intermediate for preparing chemically pure pitolisant hydrochloride, wherein the pitolisant hydrochloride obtained from this process does not contain deschloro compounds such as deschloro mesyl compound of

[0016] Formula-(IIIA) and deschloro pitolisant of Formula-(IA).

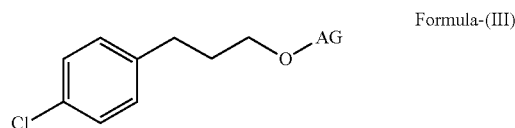
SUMMARY OF THE INVENTION

[0017] Disclosed herein is a process for preparing pitolisant and its pharmaceutically acceptable salts of Formula-(I) comprising:

[0018] a) reacting 3-(piperidin-1-yl)propan-1-ol



[0019] with Formula-(III)



[0020] in presence of a base to obtain pitolisant;

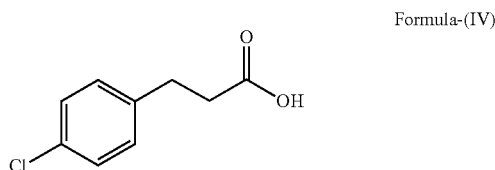
[0021] b) treating pitolisant with pharmaceutically acceptable salt; and

[0022] c) isolating pitolisant pharmaceutically acceptable salt,

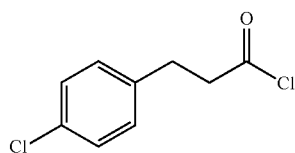
[0023] wherein the base is not sodium hydride.

[0024] Also disclosed herein is a process for preparing pitolisant and its pharmaceutically acceptable salts, polymorphs, hydrates, and solid dispersions thereof comprising:

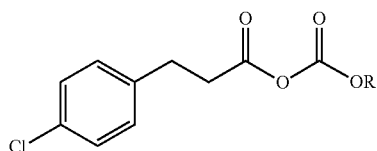
[0025] a) providing 3-(4-chlorophenyl)-propionic acid of Formula-(IV);



[0026] b) converting 3-(4-chlorophenyl)-propionic acid of Formula-(IV) to its acid chloride of Formula-(VI) or acid anhydride Formula-(VIA);

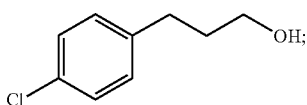


Formula-(VI)



Formula-(VIA)

[0027] c) reducing acid chloride of Formula-(VI) or acid anhydride Formula-(VIA) to obtain 3-(4-chlorophenyl)-propanol of Formula-(V)



Formula-(V)

and

[0028] d) converting 3-(4-chlorophenyl)-propanol of Formula-(V) to obtain pitolisant and its pharmaceutically acceptable salts.

[0029] The disclosure also relates to a process to obtain a solid-state form of pitolisant hydrochloride of Formula-(I) with or without isolation of crystalline pitolisant hydrochloride of Formula-(I).

DETAILED DESCRIPTION OF THE DRAWINGS

[0030] FIG. 1. Powder X-ray Diffraction (“PXRD”) pattern of solid dispersion comprising pitolisant hydrochloride and beta-cyclodextrin.

[0031] FIG. 2. PXRD of solid dispersion comprising pitolisant hydrochloride and povidone.

[0032] FIG. 3. PXRD of solid dispersion comprising pitolisant hydrochloride and copovidone.

[0033] FIG. 4. PXRD of solid dispersion comprising pitolisant hydrochloride sesquihydrate and beta-cyclodextrin.

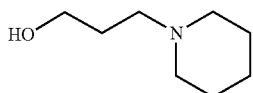
[0034] FIG. 5. PXRD of pitolisant hydrochloride sesquihydrate.

[0035] FIG. 6. PXRD of pitolisant oxalate.

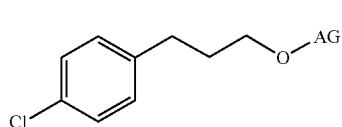
DETAILED DESCRIPTION

[0036] One embodiment relates to a process for preparing pitolisant and its pharmaceutically acceptable salts comprising:

[0037] a) reacting 3-(piperidin-1-yl)propan-1-ol



[0038] with Formula-(III)



Formula-(III)

[0039] in presence of a base to obtain pitolisant;

[0040] b) treating pitolisant with pharmaceutically acceptable salt; and

[0041] c) isolating pitolisant pharmaceutically acceptable salt,

[0042] wherein the base is not sodium hydride.

[0043] In one aspect, the activating group (AG) may be a suitable electron withdrawing moiety so that the O-AG moiety may undergo a nucleophilic substitution reaction with 3-(piperidin-1-yl)propan-1-ol or a salt thereof.

[0044] Examples of activating groups, include, but are not limited to, an alkylsulfonate (e.g., methyl sulfonate, ethylsulfonate), an arylsulfonate (e.g., phenylsulfonate, tolylsulfonate, p-nitrophenyl sulfonate, benzyl sulfonate, etc.), an alkyl ester (e.g., a trifluoromethylacetyl ester, a trichloromethylacetyl ester), an aryl ester (e.g., phenylacetyl ester, p-nitro-phenylacetyl ester, p-chloro-phenylacetyl esters), and the like.

[0045] Formula-(III) may be prepared by reacting 3-(4-chlorophenyl)propan-1-ol with a suitable activating group, as reported by, for example, Greene.

[0046] In one aspect, Formula-(III) may be selected from 3-(4-chlorophenyl)propyl methanesulfonate, 3-(4-chlorophenyl)propyl ethanesulfonate, 3-(4-chlorophenyl)propyl benzenesulfonate, 3-(4-chlorophenyl)propyl 4-methylbenzenesulfonate, 3-(4-chlorophenyl)propyl 4-nitrobenzenesulfonate, 3-(4-chlorophenyl)propyl 2,2,2-trifluoroacetate, 3-(4-chlorophenyl)propyl 2,2,2-trichloroacetate, 3-(4-chlorophenyl)propyl benzoate, 3-(4-chlorophenyl)propyl 4-nitrobenzoate, and 3-(4-chlorophenyl)propyl 4-chlorobenzoate. In yet another aspect, In another aspect, Formula-(III) may be selected from 3-(4-chlorophenyl)propyl methanesulfonate.

[0047] In one aspect, the base used in step a) is selected from sodium hydroxide, sodium carbonate, sodium bicarbonate, sodium methoxide, potassium hydroxide, potassium carbonate, potassium bicarbonate, potassium tertiary-butoxide, lithium hydroxide, lithium carbonate, lithium bicarbonate, triethylamine, t-butylamine, trimethylamine, N,N-diisopropylamine, and a combination thereof.

[0048] In another aspect, the solvent used in step a) comprises aprotic solvent. In yet another aspect, the solvent used in step a) comprises a polar, aprotic solvent.

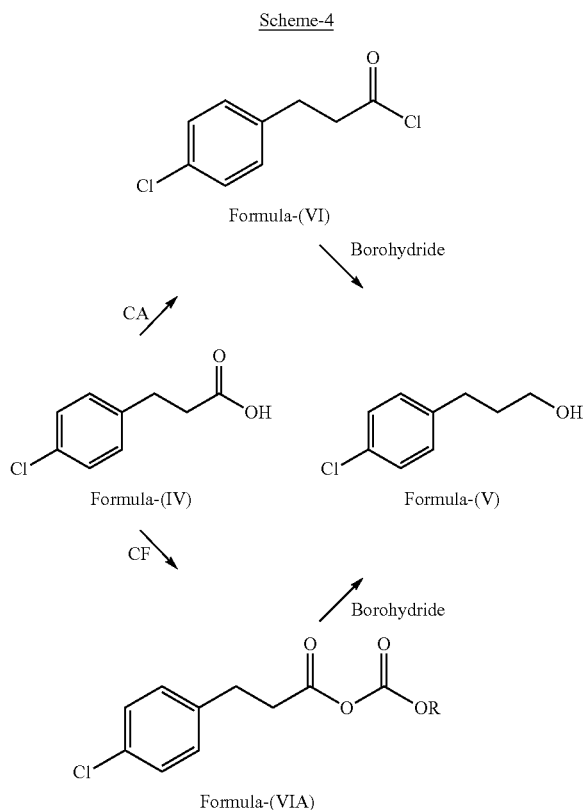
[0049] Suitable solvents (and their boiling points), include, but are not limited to N,N-dimethylformamide ($\approx 153^\circ\text{C}$.), N,N-dimethylacetamide ($\approx 165^\circ\text{C}$.), dimethylsulfoxide ($\approx 189^\circ\text{C}$.), 1-methyl-2-pyrrolidone $\approx 202^\circ\text{C}$.), 1-methyl-2-piperidone ($\approx 105\text{--}106^\circ\text{C}$. (12 mmHg), 1,3-dimethyl-2-imidazolidinone ($\approx 225^\circ\text{C}$.), acetonitrile ($\approx 82^\circ\text{C}$.), acetone ($\approx 56^\circ\text{C}$.), dichloromethane ($\approx 40^\circ\text{C}$.), ethyl acetate ($\approx 77^\circ\text{C}$.), tetrahydrofuran (66°C .), methanol ($\approx 65^\circ\text{C}$.), ethanol ($\approx 78^\circ\text{C}$.), isopropanol ($\approx 83^\circ\text{C}$.), n-butanol 118°C .), and a combination thereof.

[0050] In another aspect, the solvent used in step a) is selected from N,N-dimethylformamide, N,N-dimethylacet-

water and solvent wherein the solvent is selected from methanol, ethanol, isopropanol, butanol, dichloromethane, dimethylformamide, and dimethylsulfoxide.

[0122] The starting material particularly 3-(4-chlorophenyl)-propanol of Formula-(V) for the preparation of pitolisant hydrochloride of Formula-(I) in the present disclosure is prepared by conventional methods or methods as disclosed herein.

[0123] One embodiment relates to the preparation of 3-(4-chlorophenyl)-propanol of Formula-(V), as depicted in Scheme-4.

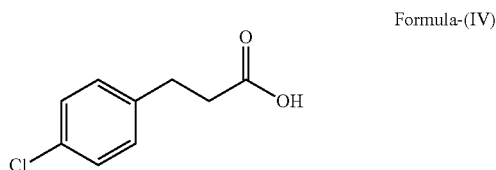


In Scheme-4, the chlorinating agent (CA) may be any suitable agent that converts the carboxylic acid moiety of Formula-(IV) to a carboxylic acid chloride of Formula-(VI). Examples of chlorinating agents include, but are not limited to, phosgene ($\text{C}(\text{O})\text{Cl}_2$), diphosgene ($\text{Cl}_3\text{COC}(\text{O})\text{Cl}$), triphosgene ($\text{Cl}_3\text{COC}(\text{O})\text{OCCl}_3$), thionyl chloride ($\text{S}(\text{O})\text{Cl}_2$), oxalyl chloride ($\text{ClC}(\text{O})-\text{C}(\text{O})\text{Cl}$), phosphorus oxychloride ($\text{P}(\text{O})\text{Cl}_3$), etc.

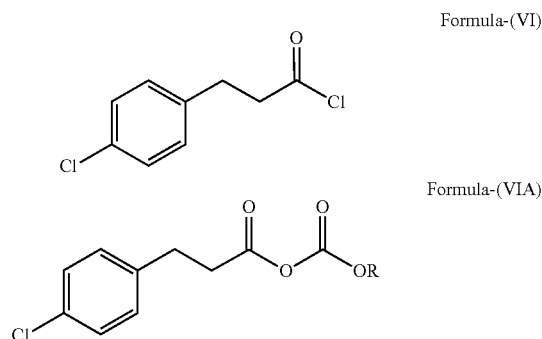
[0124] Also, in Scheme-4, the chloroformate (CF or $\text{ClC}(\text{O})\text{OR}$) may be any suitable agent that converts the carboxylic acid moiety of Formula-(IV) to the carboxylic acid anhydride of Formula-(VIA), where the R-group for Formula-(VIA) may be any suitable radical (e.g., alkyl group or phenyl), including, for example a linear or branched alkyl having 1 to 12 carbon atoms, including for example, methyl, ethyl, propyl, isopropyl, isobutyl and the like (e.g., a C_{1-6} alkyl); or in the instance of a phenyl, R may be a unsubstituted phenyl or a substituted phenyl (e.g., o-tolyl, xylyl, etc.).

[0125] In the view of the foregoing, an embodiment relates to the preparation of 3-(4-chlorophenyl)-propanol of Formula-(V), comprising

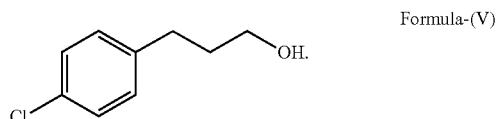
[0126] a) providing 3-(4-chlorophenyl)-propionic acid of Formula-(IV) in a solvent;



[0127] b) converting 3-(4-chlorophenyl)-propionic acid of Formula-(IV) to its acid chloride of Formula-(VI) or acid anhydride Formula-(VIA); and



[0128] c) reducing acid chloride of Formula-(VI) or acid anhydride Formula-(VIA) to obtain 3-(4-chlorophenyl)-propanol of Formula-(V)



[0129] In one aspect, the 3-(4-chlorophenyl)-propionic acid of step a) is dissolved in solvent selected from dichloromethane, dichloroethane, tetrahydrofuran, toluene, dioxane, N,N-dimethylformamide, N,N-dimethylacetamide, dimethylsulfoxide, 1-methyl-2-pyrrolidone, 1-methyl-2-piperidone, 1,3-dimethyl-2-imidazolidinone, acetonitrile, acetone, ethyl acetate, methanol, ethanol, isopropanol, n-butanol, and combination thereof.

[0130] In another aspect, the conversion of 3-(4-chlorophenyl)-propionic acid of Formula-(IV) to its acid chloride of Formula-(VI) is carried out in presence of a chlorinating agent, including, for example, phosgene ($\text{C}(\text{O})\text{Cl}_2$), diphosgene ($\text{Cl}_3\text{COC}(\text{O})\text{Cl}$), triphosgene ($\text{Cl}_3\text{COC}(\text{O})\text{OCCl}_3$), thionyl chloride ($\text{S}(\text{O})\text{Cl}_2$), oxalyl chloride ($\text{ClC}(\text{O})-\text{C}(\text{O})\text{Cl}$), phosphorus oxychloride ($\text{P}(\text{O})\text{Cl}_3$), etc.

[0131] In yet another aspect, the conversion of 3-(4-chlorophenyl)-propionic acid of Formula-(IV) to its acid anhydride of Formula-(VIA) is carried out in presence of a

haloformate selected from methylchloroformate, ethylchloroformate, isopropyl chloroformate, isobutyl chloroformate, phenyl chloroformate, etc.

[0132] In another aspect, the reaction of step b) is carried out in presence of solvent selected from dichloromethane, dichloroethane, tetrahydrofuran, toluene, dioxane, N,N-dimethylformamide, N,N-dimethylacetamide, dimethylsulfoxide, 1-methyl-2-pyrrolidone, 1-methyl-2-piperidone, 1,3-dimethyl-2-imidazolidinone, acetonitrile, acetone, ethyl acetate, methanol, ethanol isopropanol, n-butanol, and a combination thereof.

[0133] In yet another aspect, the reduction reaction of step c) is carried out in presence of a borohydride selected from lithium borohydride, sodium borohydride, potassium borohydride, sodium cyanoborohydride, lithium triethyl borohydride, and a combination thereof. In one aspect, the reduction reaction of step c) is carried out in the presence of sodium borohydride.

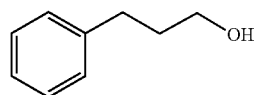
[0134] One will appreciate that the reagents may be used in an amount corresponding to a about 1-mole equivalent based on the reactant. For instance, in the conversion of 3-(4-chlorophenyl)-propionic acid of Formula-(IV) to its acid chloride of Formula-(VI) or acid anhydride Formula-(VIA), one will appreciate that the amount of chlorinating agent or chloroformate will be based on the amount of 3-(4-chlorophenyl)-propionic acid. Accordingly, reaction of about 1 mol of 3-(4-chlorophenyl)-propionic acid may require about 1 mol of chlorinating agent or about 1 mol of chloroformate. In certain instances, it may be desirable to use an excess of chlorinating agent or chloroformate, e.g., 1.05-mol excess, 1.1-mol excess, and the like. The same reasoning may be applicable with respect to a borohydride used in step c).

[0135] In another aspect, the reduction reaction of step c) is carried out in same solvent as used in step b) and in optionally presence of water.

[0136] In another aspect, the reduction reaction of step c) is carried out at a temperature in the range of 0° C. to 25° C.

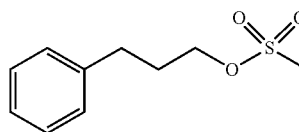
[0137] In another aspect, the acid chloride or anhydride formation and reduction reaction of steps b) and reduction step c) are carried out with or without isolating acid chloride or anhydride.

[0138] In another embodiment of the present disclosure, the pitolisant hydrochloride obtained from processes disclosed herein is substantially pure and has a chemical purity $\geq 98\%$, $\geq 98.5\%$, $\geq 99\%$, $\geq 99.5\%$, $\geq 99.6\%$, $\geq 99.7\%$, $\geq 99.8\%$, $\geq 99.9\%$, or $\geq 99.98\%$ as determined by HPLC. In another aspect of the present invention, pitolisant hydrochloride obtained from present disclosure is essentially free of deschloro impurities selected from compounds of Formula-(VA), deschloro mesyl compound of Formula-(IIIA) and deschloro pitolisant of Formula-(IA)

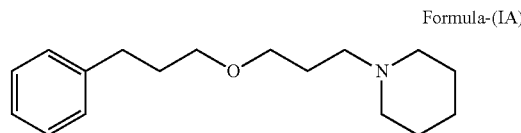


Formula-(VA)

-continued



Formula-(IIIA)



Formula-(IA)

[0139] In another embodiment of the present disclosure, the composition comprising pitolisant hydrochloride obtained from processes described herein has total impurities less than 1.0%, less than 0.5%, or less than 0.2%, based on the total weight of the composition. Still another embodiment of the present disclosure, the composition comprising pitolisant hydrochloride obtained from processes described herein has less than 1.0% of deschloro impurities selected from compounds of Formula-(VA), deschloro mesyl compound of Formula-(IIIA), and deschloro pitolisant of Formula-(IA).

[0140] HPLC purity of pitolisant hydrochloride obtained by processes disclosed herein may be greater than or equal to 98% by weight of the composition, as evidenced by the following results of a representative sample.

Assay by HPLC (% w/w)	99.6%
Related Substances by HPLC (% w/w):	
Benzyl amino alcohol ^(a)	Not detected
Benzyl piperazine ^(b)	Not detected
N-oxide ^(c)	Not detected
Des impurity ^(d)	Not detected
Highest unspecified impurity	0.03% w/w
Total impurities	0.05% w/w

Notes: ^(a)Benzyl amino alcohol (1-[3-(4-Chloro-benzyloxy)-propyl]-piperidine) ^(b)Benzyl piperazine (1-(4-Chloro-benzyl)-piperidine) ^(c)N-oxide (1-[3-(4-Chloro-phenyl)-propoxy]-piperidine 1-oxide) ^(d)Des impurity (Deschloro pitolisant)

[0141] Disclosed herein is a pharmaceutical composition (e.g., tablet, capsule, or suspension) for administration of a therapeutically effective amount of pitolisant to a patient in need thereof for the treatment of excessive daytime sleepiness (EDS) or cataplexy in adult patients with narcolepsy. In one aspect, the pharmaceutical compositions may include one or more of an excipient selected from a carrier, a filler, a diluent, a suspending agent, a lubricant, a glidant, a sweetener, a stabilizing agent, a solubilizing agent, an antioxidant, a preservative, a flavoring agent, a binder, a colorant, an osmotic agent, a buffer, a surfactant, a disintegrant, a granulating agent, and a coating material. For instance, the pharmaceutical composition (e.g., tablet) may comprise a therapeutically effective amount of pitolisant (e.g., about 4.45 mg or about 17.8 mg), an excipient, and a coating layer.

[0142] In one aspect, the pharmaceutical composition disclosed herein comprises any one of the solid-state forms of pitolisant hydrochloride disclosed herein, such as, for example, pitolisant hydrochloride dihydrate, pitolisant hydrochloride sesquihydrate (including crystalline pitolisant hydrochloride sesquihydrate), pitolisant hydrochloride

providing a solution of pitolisant hydrochloride in a solvent; b) adding one or more pharmaceutically acceptable excipients to the solvent; and c) removing the solvent from the reaction mass.

[0243] Aspect 7. The solid dispersion of Aspect 6, wherein the removal of solvent from the reaction is carried out by at least one of evaporation, spray drying, and freeze drying.

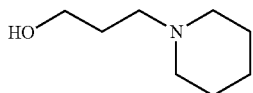
[0244] Aspect 8. A process for the preparation a solid dispersion comprising pitolisant hydrochloride and one or more pharmaceutically acceptable excipients, said process comprising: a. providing pitolisant first salt in a solvent; b. neutralizing the first pitolisant salt with a base to obtain pitolisant; c. treating the pitolisant with hydrochloride to obtain a pitolisant hydrochloride solution; d. adding one or more pharmaceutically acceptable excipients; e. removing the solvent from the reaction mass; and f. isolating a solid dispersion comprising pitolisant hydrochloride and the one or more pharmaceutical acceptable excipients; wherein the reaction is carried out without isolation of pitolisant hydrochloride salt.

[0245] Aspect 9. The process of Aspect 8, wherein the pitolisant first salt is an acid addition salt where the acid is selected from hydrobromic acid, hydroiodic acid, oxalic acid, fumaric acid, maleic acid, succinic acid, acetic acid, adipic acid, ascorbic acid, aspartic acid, benzoic acid, camphoric acid, camphor-10-sulfonic acid, capric acid (decanoic acid), caproic acid (hexanoic acid), carbonic acid, cinnamic acid, citric acid, cyclamic acid, dodecylsulfuric acid, ethane-1,2-disulfonic acid, ethanesulfonic acid, formic acid, galactaric acid, gentisic acid, glucoheptonic acid, gluconic acid, glucuronic acid, glutamic acid, glutaric acid, glycerophosphoric acid, glycolic acid, hippuric acid, isobutyric acid, lactic acid, lactobionic acid, lauric acid, maleic acid, malic acid, malonic acid, mandelic acid, naphthalene-1,5-disulfonic acid, naphthalene-2-sulfonic acid, nicotinic acid, nitric acid, oleic acid, oxalic acid, palmitic acid, pamoic acid, pyroglutamic acid, salicylic acid, sebacic acid, stearic acid, succinic acid, tartaric acid, thiocyanic acid, undecylenic acid, aminocaproic acid, caprilic acid, alpha lipoic acid, R-lipoic acid, myristic acid, myristoleic acid, palmitoleic acid, elaidic acid, linoleic acid, linolenic acid, linolelaidic acid, arachidonic acid, and a combination thereof.

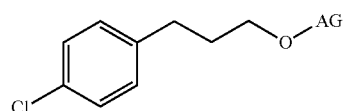
[0246] Aspect 10. The process of any one of Aspects 8-9, wherein the base in step b) comprises an inorganic base, an organic base, or a combination thereof selected from sodium hydroxide, potassium hydroxide, lithium hydroxide, sodium carbonate, potassium carbonate, lithium carbonate, sodium bicarbonate, potassium bicarbonate, lithium bicarbonate, trimethylamine, and diisopropylamine, and a combination thereof.

[0247] Aspect 11. A process for preparing pitolisant or a pharmaceutically acceptable salt thereof, comprising:

[0248] a) reacting 3-(piperidin-1-yl)propan-1-ol



[0249] with Formula-(III);



Formula-(III)

[0250] in the presence of a solvent and a base to obtain pitolisant;

[0251] b) treating pitolisant with a pharmaceutically acceptable acid; and

[0252] c) isolating the pitolisant pharmaceutically acceptable salt

[0253] wherein the base is not sodium hydride; and

[0254] wherein AG is selected from an alkylsulfonate, an arylsulfonate, an alkyl ester, and an aryl ester.

[0255] Aspect 12. The process of Aspect 11, wherein the base used in step a) is selected from sodium hydroxide, sodium carbonate, sodium bicarbonate, sodium methoxide, potassium hydroxide, potassium carbonate, potassium bicarbonate, potassium tertiary-butoxide, triethylamine, t-butylamine, trimethylamine and N, N-diisopropylamine, and a combination thereof.

[0256] Aspect 13. The process of any one of Aspects 11-12, wherein the solvent of step a) is selected from N,N-dimethylformamide, N,N-dimethylacetamide, dimethylsulfoxide, 1-methyl-2-pyrrolidone, 1-methyl-2-piperidone, 1,3-dimethyl-2-imidazolidinone, acetonitrile, acetone, dichloromethane, ethyl acetate, tetrahydrofuran, acetone, and a combination thereof.

[0257] Aspect 14. Pitolisant hydrochloride sesquihydrate.

[0258] Aspect 15. The pitolisant hydrochloride sesquihydrate of Aspect 14 having PXRD peaks at 5.3, 7.3, 10.0, 11.8, and $26.1 \pm 0.2^\circ$ two-theta.

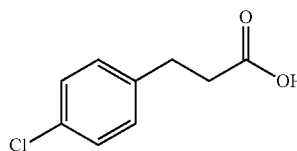
[0259] Aspect 16. Crystalline pitolisant oxalate having PXRD peaks at 7.4, 14.8, 21.8, 22.3, and $29.8 \pm 0.2^\circ$ two-theta.

[0260] Aspect 17. A process for preparing pitolisant hydrochloride, which comprises converting the crystalline pitolisant oxalate of Aspect 16 to pitolisant hydrochloride, its hydrates, and solid-state forms thereof.

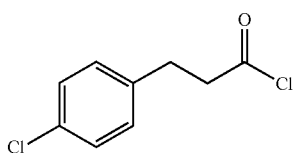
[0261] Aspect 18. A process for preparing pitolisant or a pharmaceutically acceptable salt thereof, which comprises the preparation of 3-(4-chlorophenyl)-propanol of Formula-(V), said process comprising:

[0262] a) providing 3-(4-chlorophenyl)-propionic acid of Formula-(IV);

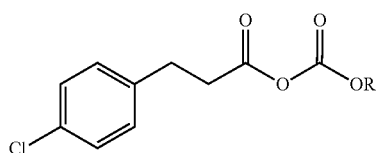
Formula-(IV)



[0263] b) converting 3-(4-chlorophenyl)-propionic acid of Formula-(IV) to its acid chloride of Formula-(VI) or acid anhydride of Formula-(VIA);



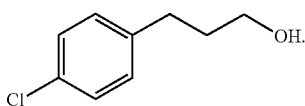
Formula-(VI)



Formula-(VIA)

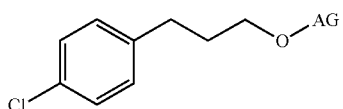
[0264] and

[0265] c) reducing acid chloride of Formula-(VI) or acid anhydride Formula-(VIA), where R is an alkyl or a phenyl, to obtain 3-(4-chlorophenyl)-propanol of Formula-(V)



Formula-(V)

[0266] Aspect 19. The process of Aspect 18, further comprising reacting 3-(4-chlorophenyl)-propanol of Formula-(V) with Formula-(III)

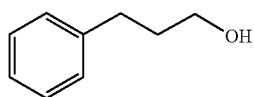


Formula-(III)

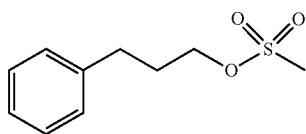
[0267] to obtain pitolisant;

[0268] wherein AG is selected from an alkylsulfonate, an arylsulfonate, an alkyl ester, and an aryl ester.

[0269] Aspect 20. Pitolisant, its pharmaceutically acceptable salt, hydrate, and solid-state form thereof having a chemical purity not less than 98% and substantially free of impurities selected from Formula-(VA), deschloro mesyl compound of Formula-(IIIA), and deschloro pitolisant of Formula-(IA)

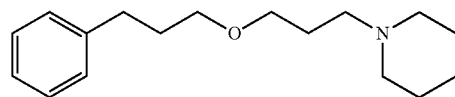


Formula-(VA)



Formula-(IIIA)

-continued



Formula-(IA)

[0270] The subject matter of Indian Patent Application Nos. 202141025328 (filed on Jun. 7, 2021), 202141035609 (filed on Aug. 6, 2021), 202141046743 (filed on Oct. 13, 2021), and 202241009416 (filed on Feb. 22, 2022) is incorporated by reference in its entirety. Additionally, the subject matter of the documents cited herein is incorporated by reference in their entirety to the extent necessary. If there is a difference in meaning between the incorporated terms and the terms disclosed herein, the meaning of the terms disclosed herein will control.

CITED INFORMATION

[0271] EFSA Panel et al., Safety of low-substituted hydroxypropyl cellulose (L-HPC) to be used as a food additive in food supplements in tablet form, EFSA Journal (2018) 16(1): 5062 (“EFSA Panel”).

[0272] European Patent Application Publication No. EP 3 239 138 A1, Hydrogen fumarate salt of 1-[3-[3-(4-chlorophenyl)propoxy]propyl]-piperidine, published on Nov. 1, 2017 to Stefinovic et al. of Sandoz AG (“Sefinovic”).

[0273] Greene et al., Protective Groups in Organic Synthesis (2nd ed., John Wiley & Sons, Inc.) (1991), pages 87-118 (“Greene”).

[0274] Handbook of Pharmaceutical Excipients, 6th Ed., Eds. Rowe et al. (2009) (“Handbook”).

[0275] International Publication No. WO 2007/006708 A1, Process for preparing 1-[3-[3-(4-chlorophenyl)propoxy]propyl]-piperidine, published on Jan. 18, 2007, to Sallarés et al. of Ferrer Internacional, S.A. (“Sallarés”).

[0276] International Publication No. WO 2021/023634 A1, Process for the synthesis of pitolisant HCl, published on Feb. 11, 2021, to Colombano et al. of Procos S.P.A. (“Colombano”).

[0277] Meier et al., Influence of imidazole replacement in different structural classes of histamine H3-receptor antagonists, Eur. J. Pharm. Sci. (2001) 13(3): 249-259 (“Meier”).

[0278] Remington's Pharmaceutical Sciences (17th ed., Mack Publishing Company) and Remington: The Science and Practice of Pharmacy (21st ed., Lippincott Williams & Wilkins), collectively “Remington's.”

[0279] Sarode et al., Low-Viscosity Hydropropylcellulose (HPC) Grades SL and SSL: Versatile Pharmaceutical Polymers for Dissolution Enhancement, Controlled Release, and Pharmaceutical Processing, AAPS PharmSciTech (2013) 14(1): 151-159 (“Sarode”).

[0280] U.S. Pat. No. 7,910,605 B2, Non-imidazole alkylamines as histamine H3-receptor ligands and their therapeutic applications, issued on Mar. 2, 2011 to Schwartz et al. of Societe Civile Bioproject (“Schwartz”).

[0281] U.S. Pat. No. 8,207,197 B2, Monohydrochloride salt of 1-[3-([3-(4-chlorophenyl)propoxy]propyl)-piperidine], issued on Jun. 26, 2012 to Raga et al. of Bioproject (“Raga”).

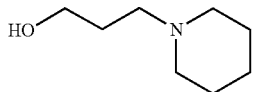
[0282] WAKIX® (pitolisant) tablets prescribing information, as of Oct. 13, 2020 (“WAKIX® Label”).

9. The process of claim 8, wherein the pitolisant first salt is an acid addition salt where the acid is selected from hydrobromic acid, hydroiodic acid, oxalic acid, fumaric acid, maleic acid, succinic acid, acetic acid, adipic acid, ascorbic acid, aspartic acid, benzoic acid, camphoric acid, camphor-10-sulfonic acid, capric acid (decanoic acid), caproic acid (hexanoic acid), carbonic acid, cinnamic acid, citric acid, cyclamic acid, dodecylsulfuric acid, ethane-1,2-disulfonic acid, ethanesulfonic acid, formic acid, galactaric acid, gentisic acid, glucoheptonic acid, gluconic acid, glucuronic acid, glutamic acid, glutaric acid, glycerophosphoric acid, glycolic acid, hippuric acid, isobutyric acid, lactic acid, lactobionic acid, lauric acid, maleic acid, malic acid, malonic acid, mandelic acid, naphthalene-1,5-disulfonic acid, naphthalene-2-sulfonic acid, nicotinic acid, nitric acid, oleic acid, oxalic acid, palmitic acid, pamoic acid, pyroglutamic acid, salicylic acid, sebacic acid, stearic acid, succinic acid, tartaric acid, thiocyanic acid, undecylenic acid, aminocaproic acid, caprylic acid, alpha lipoic acid, R-lipoic acid, myristic acid, myristoleic acid, palmitoleic acid, elaidic acid, linoleic acid, linolenic acid, linolelaidic acid, arachidonic acid, and a combination thereof.

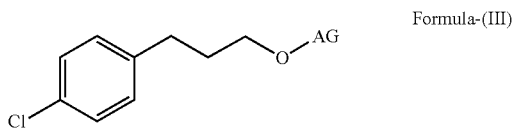
10. The process of claim 8, wherein the base in step b) comprises an inorganic base, an organic base, or a combination thereof selected from sodium hydroxide, potassium hydroxide, lithium hydroxide, sodium carbonate, potassium carbonate, lithium carbonate, sodium bicarbonate, potassium bicarbonate, lithium bicarbonate, trimethylamine, and diisopropylamine, and a combination thereof.

11. A process for preparing pitolisant or a pharmaceutically acceptable salt thereof, comprising:

a) reacting 3-(piperidin-1-yl)propan-1-ol



with Formula-(III);



in the presence of a solvent and a base to obtain pitolisant;

b) treating pitolisant with a pharmaceutically acceptable acid; and

c) isolating the pitolisant pharmaceutically acceptable salt;

wherein the base is not sodium hydride; and

wherein AG is selected from an alkylsulfonate, an arylsulfonate, an alkyl ester, and an aryl ester.

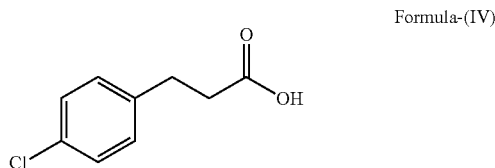
12. The process of claim 11, wherein the base used in step a) is selected from sodium hydroxide, sodium carbonate, sodium bicarbonate, sodium methoxide, potassium hydroxide, potassium carbonate, potassium bicarbonate, potassium tertiary-butoxide, triethylamine, t-butylamine, trimethylamine and N, N-diisopropylamine, and a combination thereof.

13. The process of claim 11, wherein the solvent of step a) is selected from N,N-dimethylformamide, N,N-dimethyl-

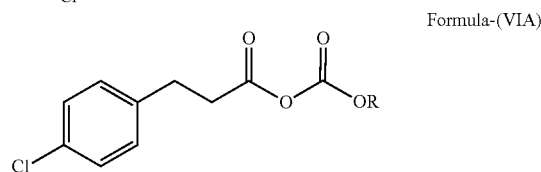
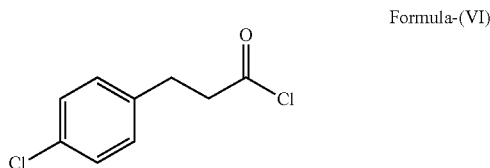
acetamide, dimethylsulfoxide, 1-methyl-2-pyrrolidone, 1-methyl-2-piperidone, 1,3-dimethyl-2-imidazolidinone, acetonitrile, acetone, dichloromethane, ethyl acetate, tetrahydrofuran, acetone, and a combination thereof.

14. The process of claim 11, further comprising preparing 3-(4-chlorophenyl)propanol of Formula-(V)

a) providing 3-(4-chlorophenyl)propionic acid of Formula-(IV);

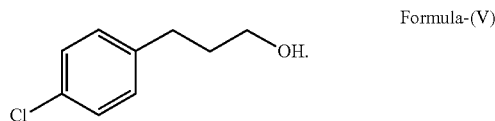


b) converting 3-(4-chlorophenyl)propionic acid of Formula-(IV) to its acid chloride of Formula-(VI) or acid anhydride of Formula-(VIA);

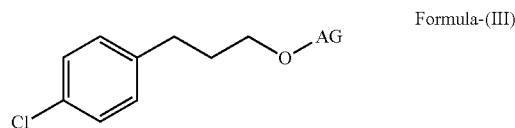


and

c) reducing acid chloride of Formula-(VI) or acid anhydride Formula-(VIA), where R is an alkyl or a phenyl, to obtain 3-(4-chlorophenyl)propanol of Formula-(V)



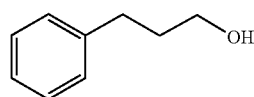
15. The process of claim 14, further comprising reacting 3-(4-chlorophenyl)propanol of Formula-(V) with Formula-(III)



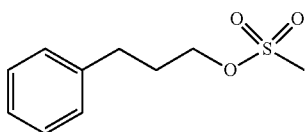
to obtain pitolisant;

wherein AG is selected from an alkylsulfonate, an arylsulfonate, an alkyl ester, and an aryl ester.

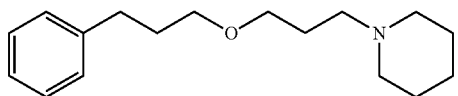
16. The process of claim **14**, wherein the pitolisant or it's a pharmaceutically acceptable salt thereof has a chemical purity not less than 98% and substantially free of impurities selected from Formula-(VA), deschloro mesyl compound of Formula-(IIIA), and deschloro pitolisant of Formula-(IA)



Formula-(VA)



Formula-(IIIA)



Formula-(IA)

17. Pitolisant hydrochloride sesquihydrate.

18. The pitolisant hydrochloride sesquihydrate of claim **17** having PXRD peaks at 5.3, 7.3, 10.0, 11.8, and $26.1 \pm 0.2^\circ$ two-theta.

19. The pitolisant hydrochloride sesquihydrate of claim **18** having one or more PXRD peaks at 2.6, 3.6, 8.8, 10.8,

11.1, 11.5, 13.6, 14.0, 14.4, 14.8, 15.3, 16.2, 16.6, 17.3, 17.6, 18.3, 18.9, 19.5, 19.8, 20.6, 21.6, 21.9, 24.0, 24.9, 25.4, 27.1, 28.8, 29.1, 30.1, 30.7, 33.8, 33.9, 34.9, and $35.5 \pm 0.2^\circ$ two-theta.

20. The pitolisant hydrochloride sesquihydrate of claim **18** having four or more PXRD peaks at 2.6, 3.6, 8.8, 10.8, 11.1, 11.5, 13.6, 14.0, 14.4, 14.8, 15.3, 16.2, 16.6, 17.3, 17.6, 18.3, 18.9, 19.5, 19.8, 20.6, 21.6, 21.9, 24.0, 24.9, 25.4, 27.1, 28.8, 29.1, 30.1, 30.7, 33.8, 33.9, 34.9, and $35.5 \pm 0.2^\circ$ two-theta.

21. The pitolisant hydrochloride sesquihydrate of claim **18** having a PXRD pattern substantially as shown in FIG. **5**.

22. Crystalline pitolisant oxalate having PXRD peaks at 7.4, 14.8, 21.8, 22.3, and $29.8 \pm 0.2^\circ$ two-theta.

23. The crystalline pitolisant oxalate of claim **22** having one or more PXRD peaks at 12.3, 15.8, 17.0, 17.9, 18.8, 20.1, 20.4, 21.0, 23.4, 23.9, 24.3, 24.7, 25.2, and $35.7 \pm 0.2^\circ$ two-theta.

24. The crystalline pitolisant oxalate of claim **22** having two or more PXRD peaks at 12.3, 15.8, 17.0, 17.9, 18.8, 20.1, 20.4, 21.0, 23.4, 23.9, 24.3, 24.7, 25.2, and $35.7 \pm 0.2^\circ$ two-theta.

25. The crystalline pitolisant oxalate of claim **22** having a PXRD pattern substantially as shown in FIG. **6**.

26. A process for preparing pitolisant hydrochloride, which comprises converting the crystalline pitolisant oxalate of claim **16** to pitolisant hydrochloride, its hydrates, and solid-state forms thereof.

* * * * *